

Beyond the Symmetric Thin Ship: Asymmetry, Centreplane Lifting, and Heel

(Part II of the `michell` calculation notes)

`michell`

Abstract

Part I documented the classical Michell wave-resistance integral for a hull that is symmetric about its centerplane: a thin *thickness* distribution radiating a *source* wave system, evaluated exactly per B-spline knot span. This second note documents the three extensions built on top of that kernel, all of which reuse the Part I machinery essentially unchanged. First, a *port/starboard-asymmetric* hull is decomposed into a symmetric thickness part and an antisymmetric *camber* part; the camber radiates a centreplane *y-dipole* wave system that superposes on the source system without a cross term. Second, the dipole *magnitude* — which no pointwise boundary condition fixes, because the doublet density is a non-local functional of the camber slope — is obtained honestly by *solving* a centreplane lifting-surface problem with a horseshoe vortex lattice, replacing the crude strip closure $\mu = 2Uf_a$. Third, a *heeled* hull is handled by tilting the source centreplane, which turns the vertical decay rate *complex* and otherwise leaves the exact per-span kernel untouched. Throughout, section references point at the Rust source. All notation and constants are inherited from Part I; in particular $\nu = g/U^2$, $k_x = \nu\lambda$, $\kappa = \nu\lambda^2$, and $F(\lambda) = I + iJ$ is the source free-wave amplitude of Part I, Eq. (inner).

Contents

1	Why the symmetric integral is not enough	2
2	Decomposition of an asymmetric hull	2
3	The centreplane dipole wave system	3
3.1	Source plus dipole	3
3.2	The strip closure and the real dipole weight	3
3.3	Additive resistance and the θ -parity argument	4
4	The centreplane lifting solve (“level 2”)	4
4.1	Two-dimensional proving ground	4
4.2	Three-dimensional horseshoe vortex lattice	5
4.3	Free surface as a rigid wall (double body)	5
4.4	From the solved density to the wave amplitude	6
5	Heeled hulls	6
6	Data-flow summary	7

1 Why the symmetric integral is not enough

The classical Michell integral of Part I represents the hull by a single-valued half-breadth $f(x, z) \geq 0$ and a sheet of *sources* of strength $\sigma = 2U \partial f / \partial x$ on the centerplane. That model is blind to three physically important situations:

1. **Port/starboard asymmetry.** A hull whose two sides differ — a cambered demihull, an asymmetric catamaran float, a hull at a small drift (yaw) angle — carries a transverse *lift* in addition to its thickness. Its wave system has an antisymmetric component that a single half-breadth cannot express.
2. **The lifting closure.** The strength of the antisymmetric (dipole) system is *not* fixed pointwise by the body boundary condition, unlike the source strength. Getting its magnitude right requires solving a lifting-surface integral equation.
3. **Heel.** A hull rolled about its longitudinal axis is asymmetric *relative to the horizontal free surface*, even if it is geometrically symmetric about its own centreplane. Its keel swings off the earth-vertical plane and it can no longer be written as port/starboard half-beams there.

The unifying idea is that all three are small perturbations that enter through the *same* inner integral and the *same* outer θ -quadrature as Part I. The only new ingredients are (a) a second coefficient field for the antisymmetric slope, (b) a real spectral *weight* coupling that field to a transverse-wavenumber factor, (c) a linear solve for the lifting density, and (d) a complex generalisation of the vertical-decay moment. Table 1 maps each extension to its entry point in the crate.

Extension	Entry point	Kernel change
Asymmetry (strip closure)	<code>multihull_wave_resistance_with</code>	second slope field $\partial f_a / \partial x$
Asymmetry (solved lift)	<code>asymmetric_wave_resistance_lifting</code> <code>multihull_wave_resistance_lifting</code>	centreplane vortex lattice
Heel	<code>heel_wave_resistance</code>	complex vertical decay κ

Table 1: The three extensions and where they attach to the Part I kernel (`michell.rs`).

2 Decomposition of an asymmetric hull

An asymmetric hull is built from *two* half-breadth surfaces (`Hull::new_asymmetric`, `hull.rs`): a starboard side $y = +f_+(x, z) \geq 0$ and a port side $y = -f_-(x, z) \leq 0$, both stored as non-negative magnitudes. The two surfaces must share identical degrees and knot vectors — only their control nets may differ — so that both live on the same grid of knot spans. On that shared parametrisation the hull splits linearly into a *symmetric* (thickness) part and an *antisymmetric* (camber) part,

$$f_{\text{sym}}(x, z) = \frac{1}{2}(f_+ + f_-), \quad f_a(x, z) = \frac{1}{2}(f_+ - f_-). \quad (1)$$

Because a B-spline is linear in its control net, (1) is applied directly to the control points: $P_{ij}^{\text{sym}} = \frac{1}{2}(P_{ij}^+ + P_{ij}^-)$ and $P_{ij}^a = \frac{1}{2}(P_{ij}^+ - P_{ij}^-)$. The symmetric net is itself a valid non-negative half-breadth (a mean of two non-negative nets), so the base hull is constructed from it and *every* validated

symmetric quantity of Part I — volume, LCB/VCB, the source slope field c_{ab} — is reused verbatim. The antisymmetric net produces a second per-span slope field

$$\frac{\partial f_a}{\partial x}(x, z) = \sum_{a,b} c_{ab}^a (x - x_0)^a (z - z_0)^b, \quad (\text{fx_a_coeff}), \quad (2)$$

computed by exactly the same `compute_fx_coeff` routine as the source field, and stored with the identical flattened layout. When $f_+ \equiv f_-$ the antisymmetric net is zero and the hull is bit-for-bit the symmetric `Hull::new` of Part I.

Two integrated quantities do *not* reduce to twice the mean and are corrected for the two sides separately: the wetted surface area and the centreplane transverse (waterplane) inertia both see f_+ and f_- individually. Volume, LCB/VCB, and waterplane area depend only on the sum $f_+ + f_- = 2f_{\text{sym}}$ and are already correct from the base hull.

3 The centreplane dipole wave system

3.1 Source plus dipole

The symmetric part f_{sym} radiates the classical source system with the Part I free-wave amplitude

$$F(\lambda) = \iint \frac{\partial f_{\text{sym}}}{\partial x} e^{-\kappa z} e^{ik_x x} dx dz, \quad k_x = \nu\lambda, \quad \kappa = \nu\lambda^2. \quad (3)$$

The antisymmetric part f_a is a *camber* distribution on the centreplane. Physically it is a sheet of transverse (y -normal) *doublets* of density $\mu(x, z)$: the potential jump across the centreplane. Such a doublet sheet radiates a free wave carrying *one extra factor of the transverse wavenumber* $k_y = \nu\lambda\sqrt{\lambda^2 - 1}$ relative to a source of the same integrated strength,

$$A_d(\theta) = i k_y \iint \mu(x, z) e^{-\kappa z} e^{ik_x x} dx dz. \quad (4)$$

Everything downstream is a matter of fixing μ and reducing (4) to a weight on a familiar integral.

3.2 The strip closure and the real dipole weight

The crudest way to fix the density is the *strip closure* $\mu = 2U f_a$ — the direct antisymmetric analogue of Michell's source strength $\sigma = 2U \partial f_{\text{sym}} / \partial x$. Substituting it into (4) and integrating by parts in x , using that f_a closes (vanishes) at bow and stern so the boundary term drops,

$$\iint f_a e^{-\kappa z} e^{ik_x x} dx dz = \frac{i}{\nu\lambda} \underbrace{\iint \frac{\partial f_a}{\partial x} e^{-\kappa z} e^{ik_x x} dx dz}_{G(\lambda)}, \quad (5)$$

which introduces the companion amplitude $G(\lambda)$ — the *same* inner integral as the source, but over the antisymmetric slope field (2). The two factors of i (one from the dipole, one from the integration by parts) collapse to a *real* weight in this crate's amplitude units (dividing by $2U$),

$$\boxed{A_d(\lambda) = -w(\lambda) G(\lambda), \quad w(\lambda) = C_{\text{dip}} \sqrt{\lambda^2 - 1}.} \quad (6)$$

This is `dipole_weight` in the source; the closure constant $C_{\text{dip}} = 1$ (`DIPOLE_WEIGHT_C`) is the value self-consistent with the source normalisation under the strip closure. The weight vanishes as $\lambda \rightarrow 1$ ($\theta \rightarrow 0$): a wave running dead ahead carries no transverse wavenumber and is blind to side-to-side asymmetry, so the whole dipole contribution is fed by the diverging (large- λ) part of the spectrum.

Warning on magnitude. The *structure* of (6) — the $\sqrt{\lambda^2 - 1}$ weight, the odd-in- θ parity, the additive separation below — is exact. The overall *constant* C_{dip} is not. Unlike the source strength, which the boundary condition fixes pointwise (a source sheet’s normal-velocity *jump* equals its local strength), the doublet density is fixed by the *mean* of the two-sided normal velocities, and a doublet sheet’s mean normal velocity is a hypersingular (finite-part) integral of μ . So μ is genuinely *non-local* in $\partial f_a / \partial x$: no exact pointwise closure exists, and $\mu = 2U f_a$ is only the crudest strip estimate. Getting the magnitude right is the subject of Section 4.

3.3 Additive resistance and the θ -parity argument

Combining the two systems, hull member j (at fleet offset Δx_j , transverse position y_j) contributes to the $\pm\theta$ wave systems the amplitudes

$$A_{\pm,j} = (F_j \mp w G_j) \exp[i(k_x \Delta x_j \pm k_y y_j)], \quad (7)$$

and the resistance integrand is the mean of the two signed systems, exactly as for a symmetric fleet in Part I:

$$R_w = \frac{4\rho g^2}{\pi U^2} \int_0^{\pi/2} \frac{1}{2} (|A_+|^2 + |A_-|^2) \sec^3 \theta \, d\theta, \quad (8)$$

with $A_{\pm} = \sum_j A_{\pm,j}$ (`amp_sq` in `multihull_wave_resistance_with`). The source amplitude F is *even* in θ and the dipole term wG is *odd* (the $+\theta$ system carries $-wG$, the $-\theta$ system $+wG$). Their cross term therefore integrates to zero over the symmetric Kelvin fan, and for a single hull (8) separates cleanly:

$$R_w = R_{\text{source}}(f_{\text{sym}}) + R_{\text{dipole}}(f_a). \quad (9)$$

A symmetric hull ($f_a \equiv 0$) recovers classical Michell exactly. For a fleet the dipole amplitudes superpose and interfere with the source amplitudes through the placement phases, so an asymmetric-demihull catamaran’s asymmetry and its demihull interference are handled together in one spectrum.

Kernel cost. The source and dipole amplitudes share a single pass of the z -moments (Part I, Eqs. (N)): only the x -span accumulation is repeated against the second coefficient array. `InnerIntegral::eval_pair` returns (F, G) together, so an asymmetric evaluation costs barely more than a symmetric one.

4 The centreplane lifting solve (“level 2”)

To replace the strip constant with a physically grounded density we *solve* the lifting-surface problem for $\mu(x, z)$. The centreplane $x \in [0, L]$, $z \in [0, T]$ carries the camber surface f_a , whose streamwise slope $\partial f_a / \partial x$ is the downwash the transverse doublet sheet must produce — the direct analogue of a wing’s camber-slope condition.

4.1 Two-dimensional proving ground

Before the delicate 3D free-surface solve, the crate builds and validates its 2D ancestor — classical thin-airfoil theory (`lifting.rs`) — which exercises the same machinery: discretising a lifting sheet,

imposing the Kutta condition, and recovering the loading. A thin airfoil of unit chord with camber-line slope $d\eta/dx$ is represented by a bound-vortex sheet $\gamma(x)$ satisfying the Cauchy singular integral equation

$$\frac{1}{2\pi} \text{p.v.} \int_0^1 \frac{\gamma(\xi)}{x - \xi} d\xi = U \left(\alpha - \frac{d\eta}{dx}(x) \right). \quad (10)$$

Two independent solvers cross-check each other and closed-form results:

- **Glauert Fourier series** (`glauert`), with $x = \frac{1}{2}(1 - \cos \theta)$ and $\gamma/U = 2[A_0 \frac{1+\cos \theta}{\sin \theta} + \sum_{k \geq 1} A_k \sin k\theta]$; the $\sin k\theta$ terms vanish at both edges so the Kutta condition is built in, and $C_L = \pi(2A_0 + A_1)$. A flat plate gives $A_0 = \alpha$, hence the canonical $C_L = 2\pi\alpha$.
- **Lumped-vortex lattice** (`solve_vortex_lattice`) with the Pistolesi $\frac{1}{4}-\frac{3}{4}$ rule: a point vortex at each panel's quarter-point, flow tangency collocated at its three-quarter point, which enforces the Kutta condition automatically. This is the desingularised 2D form of the hyper-singular kernel the 3D solve must integrate.

In hull terms the camber line is the antisymmetric half-beam f_a at a waterline strip and the forcing is $-U \partial f_a / \partial x$ (no separate incidence). The solved loading integrates to the doublet density $\mu(x) = \int_0^x \gamma d\xi$ — exactly the quantity the free-wave amplitude carries.

4.2 Three-dimensional horseshoe vortex lattice

The 3D engine (`lifting3d.rs`) generalises the lattice: each panel carries a *horseshoe vortex* — a bound segment at the quarter-chord plus two trailing legs running downstream to $+\infty$ along $\hat{x}$ — and flow tangency is collocated at the three-quarter-chord point. The trailing legs make the Kutta condition automatic. Assembling the normal-wash influence coefficients $A_{ij} = (\mathbf{v}_{j \rightarrow i} \cdot \hat{n}_i)$ from the Biot–Savart law (finite-segment and semi-infinite closed forms, after Katz & Plotkin) and solving

$$A \Gamma = b, \quad b_i = -\mathbf{U}_\infty \cdot \hat{n}_i, \quad (11)$$

gives the bound circulation Γ on every panel. Lift follows from Kutta–Joukowski, $C_L = 2 \sum_j \Gamma_j \Delta y_j / S$. The engine is validated on a flat rectangular wing against the two exact asymptotic limits that bracket the lift-curve slope: $dC_L/d\alpha \rightarrow 2\pi$ as $\text{AR} \rightarrow \infty$ (2D limit) and $\rightarrow \frac{\pi}{2} \text{AR}$ as $\text{AR} \rightarrow 0$ (R.T. Jones slender-wing limit), with the Prandtl lifting-line estimate $2\pi/(1 + 2/\text{AR})$ in between.

4.3 Free surface as a rigid wall (double body)

The hull-specific step (`centerplane.rs`) applies this lattice to the *vertical* centreplane. In the low-Froude near field the free surface $z = 0$ behaves as a rigid wall ($\partial\phi/\partial z = 0$), consistent with how Michell's source strength is set on the double-body flow. This is enforced by **reflecting the centreplane about** $z = 0$ with the same circulation sign, so the bound vortices run continuously through the waterline. The consequences:

- the waterline acts as a wing *root* (maximum loading) and the keel $z = T$ as a free *tip* (loading relieved to zero);
- a surface-piercing lifting surface of draft T therefore behaves as a wing of span $2T$ — the classical *effective-aspect-ratio doubling*, checked in the tests against a full wing of span $2T$ to $\sim 0.1\%$ at $\alpha = 3^\circ$.

The camber enters through the panel *normal* $\hat{n} = (-f_{a,x}, 1, 0)/\sqrt{1 + f_{a,x}^2}$, so the freestream stays $(1, 0, 0)$ and the solve is called at zero incidence. Solving for Γ and accumulating from the leading edge gives the doublet density $\mu(x, z) = \int \gamma$ on the physical half $z \geq 0$, together with a side-force coefficient $C_Y = 2 \sum_{\text{phys}} \Gamma \Delta z / (LT)$.

4.4 From the solved density to the wave amplitude

The solved density feeds the dipole free-wave amplitude by midpoint quadrature over the panel grid (`doublet_free_wave_amplitude`, each sample carrying cell area $\Delta x \Delta z$),

$$G_\mu(\lambda) = \iint \frac{\mu(x, z)}{U} e^{-\kappa z} e^{ik_x x} dx dz, \quad (12)$$

from which the dipole amplitude is

$$A_d(\lambda) = i \frac{1}{2} k_y G_\mu(\lambda), \quad k_y = \nu \lambda \sqrt{\lambda^2 - 1}. \quad (13)$$

Note that here G_μ integrates μ *directly* (not the slope), so no integration by parts is applied and the amplitude stays imaginary — contrast the strip form (6), where the by-parts step made it real. The two paths share *one* normalisation: substituting the strip closure $\mu/U = 2f_a$ into (13) and integrating by parts via (5) reproduces $A_d = -\sqrt{\lambda^2 - 1} G$ exactly. The factor $\frac{1}{2}$ is precisely what makes the closure $\mu/U = 2f_a$ collapse onto the $C_{\text{dip}} = 1$ weight. The solved μ has a different chordwise/vertical *shape* than $2f_a$, giving a dipole of the same order but a speed-dependent ratio; for a 2D flat plate $\mu_{\text{lift}}/\mu_{\text{strip}} = \pi/2$.

The single-hull entry point `asymmetric_wave_resistance_lifting` and the fleet form `multihull_wave_resistance_lifting` solve each asymmetric member's centreplane *once* (up front, not per λ), then superpose the source amplitude F_j and the solved dipole $A_{d,j}$ with the member's placement phase, exactly as in (8). The dipole phase is re-centred to the hull midpoint (the solver works on $x \in [0, L]$) so it shares the source's origin. This is the far-field coupling: each member's lifting problem is solved independently with its own free-surface image, and their dipole *wave* systems interfere through the superposition — so an asymmetric-demihull catamaran's demihull interference is captured. Near-field lifting cross-induction between close demihulls is *not* (that would need one coupled lifting solve over all centreplanes at once). The default grid is $n_x = 48$ streamwise $\times$ $n_z = 16$ vertical panels (`LiftingGrid`).

5 Heeled hulls

A hull heeled by an angle φ about its longitudinal axis is asymmetric relative to the horizontal free surface, but its keel swings off the earth-vertical centreplane, so it cannot be written as port/starboard half-beams there. Thin-ship theory instead keeps the sources on the ship's *own* (now tilted) centreplane: a strip at ship-depth z sits at earth depth $z \cos \varphi$ and transverse offset $-z \sin \varphi$. The transverse offset contributes a transverse-wavenumber phase $e^{-ik_y (-z \sin \varphi)}$ to the kernel, which combines with the earth-vertical decay $e^{-\nu \lambda^2 z \cos \varphi}$ into a single **complex vertical decay**

$$\kappa = \nu \lambda^2 \cos \varphi + i \nu \lambda \sqrt{\lambda^2 - 1} \sin \varphi = \nu \lambda^2 \cos \varphi + i k_y \sin \varphi. \quad (14)$$

The imaginary part is the same transverse-wavenumber coupling as the asymmetric dipole, arising here purely from geometry.

This is the *only* change to the Part I kernel. The x -oscillation moments M_a and the per-span accumulation are identical; only the z -moment $N_b = \int_0^{h_z} Z^b e^{-\kappa Z} dZ$ becomes complex, evaluated by `exp_moments_complex` — the *same* series and integration-by-parts recurrence as the real routine (Part I, Eqs. (Nrec)–(Nseries)), promoted to $\mathbb{C}$ with $\text{Re } \kappa \geq 0$ so the decay never grows. The real path is left entirely untouched, so $\varphi = 0$ reproduces `wave_resistance` exactly (`HeelInner` in `michell.rs`).

The tilt makes the amplitude F depend on the *sign* of θ (through the $\pm$ in $k_y \sin \varphi$), so the resistance averages the two signed systems,

$$R_w = \frac{4\rho g^2}{\pi U^2} \int_0^{\pi/2} \frac{1}{2} (|F_+|^2 + |F_-|^2) \sec^3 \theta d\theta, \quad (15)$$

where $F_{\pm}$ uses κ with $\pm i k_y \sin \varphi$. The result is *even* in φ : port and starboard heel are mirror images. Because the tilt spreads the centreplane transversely by up to $T|\sin \varphi|$, the outer-quadrature panels are sized with an effective transverse half-extent $y_{1/2} = T|\sin \varphi|$ to resolve the heel-induced oscillation.

This captures the asymmetric *wave-making* of the tilted thickness distribution — the leading heel effect. It does *not* include the lifting side-force that a heeled-and-yawed (drifting) hull develops; that would enter as a separate forcing into the centreplane lifting solve of Section 4.

6 Data-flow summary

Asymmetry — strip closure		
two nets $P_{ij}^{\pm}$	$\rightarrow$	f_{sym}, f_a (Eq. 1); slope fields c_{ab}, c_{ab}^a
c_{ab}, c_{ab}^a	$\rightarrow$	(F, G) together (<code>eval_pair</code>)
G , weight $w = \sqrt{\lambda^2 - 1}$	$\rightarrow$	dipole $A_d = -wG$ (Eq. 6)
$\frac{1}{2}(F-wG ^2 + F+wG ^2)$	$\rightarrow$	outer θ -quadrature $\Rightarrow R_w = R_{\text{src}} + R_{\text{dip}}$
Asymmetry — solved lift		
$\partial f_a / \partial x$ on centreplane	$\rightarrow$	vortex-lattice solve Γ (double body)
Γ	$\rightarrow$	doublet density $\mu = \int \gamma$, side force C_Y
μ	$\rightarrow$	G_{μ} ; dipole $A_d = i \frac{1}{2} k_y G_{\mu}$ (Eq. 13)
$F_j + A_{d,j}$, placements	$\rightarrow$	outer θ -quadrature $\Rightarrow R_w$
Heel		
heel φ	$\rightarrow$	complex κ (Eq. 14)
κ	$\rightarrow$	complex z -moments N_b (<code>exp_moments_complex</code>)
$\frac{1}{2}(F_+ ^2 + F_- ^2)$	$\rightarrow$	outer θ -quadrature $\Rightarrow R_w$ (even in φ)

All three extensions share the exact per-span inner kernel and the single smooth outer θ -quadrature of Part I. The only approximations added are the strip closure constant C_{dip} (Section 3.2, removed by the lifting solve), the panel discretisation of the lifting solve and its midpoint quadrature (Section 4), and — as in Part I — the one-dimensional angular quadrature. The wave-pattern (wake) branch of Part I is unchanged: the free-wave elevation grid is still built from the symmetric source amplitudes.

References. Tuck (1989); Dambrine, Pierre & Rousseaux (2016); Tuck, Scullen & Lazauskas (2001, 2002); Katz & Plotkin, *Low-Speed Aerodynamics* (horseshoe-lattice VORTXL); Glauert, *The*

Elements of Aerofoil and Airscrew Theory; Kaklis & Papanikolaou (21st Symp. Naval Hydro., Appendix A, on the hypersingular doublet closure).